

I .Single Choice (24 points)

- 1、 Which of following translates a program written in a high-level language into machine code?
 - A. a mouse
 - B. a compiler
 - C. an operating system
 - D. an editor
- 2、 One of the following statements does not show a proper use of the *sqrt* library function. Which one is the wrong statement? (Assume all variables are *float* variables.)
 - A. `y = sqrt(x);`
 - B. `y = 3.85 * sqrt(x + 4.2);`
 - C. `cout << sqrt(x);`
 - D. `sqrt(25.0*x);`
- 3、 Which of the following statements sends a newline character to the standard output device?
 - A. `cout << endl;`
 - B. `cout << '\n';`
 - C. `cout << \n;`
 - D. a and b above
- 4、 If the int variables i, j, and k contain the values 10, 3, and 20, respectively, what is the value of the following logical expression: `j < 4 || j == 5 && i <= k`
 - A. 3
 - B. false
 - C. true
 - D. 20
- 5、 In C++, a function prototype is
 - A. a declaration but not a definition.
 - B. a definition but not a declaration.
 - C. both a declaration and a definition.
 - D. neither a declaration nor a definition.
- 6、 Given the declarations
 - `enum MovieRatings {G, PG, PG13, R, X};`
 - `MovieRatings thisOne;`and assuming that `thisOne` currently contains a value less than `X`, which of the following statements can be used to "increment" `thisOne`?
 - A. `thisOne = thisOne + 1;`
 - B. `thisOne++;`
 - C. `thisOne = MovieRatings(thisOne + 1);`
 - D. `MovieRatings(thisOne++);`
- 7、 If a C++ class declaration begins as
 - `class A : public B`
 - `{`then which of the following is true?
 - A. Public members of B become public members of A.
 - B. Public members of B become private members of A.
 - C. Public members of A become public members of B.
 - D. Public members of A become private members of B.
- 8、 If the following function is called with a value of 3 for `n`, what is the resulting output?
 - `void TapDance( /* in */ int n )`
 - `{`
 - `if (n > -1)`
 - `{`
 - `TapDance(n - 1);`
 - `cout << "Flap";`
 - `}`
 - `else`
 - `cout << "Shuffle";`
 - `}`
 - A. FlapFlapFlapFlapShuffle
 - B. ShuffleFlapFlapFlapFlap
 - C. ShuffleFlapFlap
 - D. FlapFlapShuffle

II .What is the output of the following C++ program? (40 points)

1、 What is the output of the following program? (5 points)

```
#include <iostream.h>
#include <iomanip>
using namespace std;

int main(void) {
double x = 7 / 3.0;
double y = 7 / 3;
double z = 7 % 3;

cout<<showpoint<<fixed<<setprecision(3);
cout << "x = " << x << " ";
cout.width(5);
cout << "y = " << y << " ";
cout << "z = " << z << " " << endl;
return 0;
}
```

2、 What is the output of the following program? (5 points)

```
#include <iostream>
using namespace std;

int main()
{
    int a=6;
    a++;
    cout << "The number is: ";
    switch(a%2)
    {
    case 1:
        cout << "one ";
    case 2:
        cout << "two ";
    case 3:
        cout << "three ";
    case 4:
        cout << "four ";
    }
```

```
        break;
    case 5:
        cout << "five ";
    default:
        cout << "?";
    }
    cout << endl;
    return 0;
}
```

3、 What is the output of the following program? (8 points)

```
#include <iostream.h>
void Swap( int &a, int &b);
int main()
{
    int x( 10 ), y( 7 );
    cout<<"x="<<x<<"    y="<<y<<endl;
    Swap( x , y );
    cout<<"x="<<x<<"    y="<<y<<endl;
}
```

```
void Swap(int &a, int &b)
{
    int temp;
    temp = a ;
    a=b ;
    b=temp ;
}
```

4、 What is the output of the following program? (5 points)

```
#include <iostream.h>

using namespace std;

void SubProg();
```

```

int main()
{
    int i;
    for(i=0;i<3;i++)
    {
        SubProg();
    }
    return 0;
}
void SubProg()
{
    int i=0;
    static int j=0;

    i++;
    j++;
    cout<<i<<' '<<j<<endl;
}

```

5、 What is the output of the following program? (5 points)

```

#include <iostream>

using namespace std;

int main(){
    char cstr1[] = {"Hello"};
    char cstr2[] = {"C-+"};
    char * pCstr2 = cstr2;
    cout << cstr1 << ',' << cstr1[1] << ',' << *(cstr1+4) << ',';
    cout << *pCstr2; pCstr2 += 2;
    cout << *pCstr2 << pCstr2 - cstr2;
    return 0;
}

```

6、 What is the output of the following program? (6points)

```

#include <iostream>
using namespace std;
class B{
public:
    virtual void show() {cout<<"B";}
};
class D:public B{
public:
    void show() {cout<<"D"; }
};
void fun1(B &ptr)
{    ptr.show();
}
void fun2(B ptr2)
{    ptr2.show();
}
int main()
{    B b;
    D d;
    fun1(b);
    fun1(d);
    fun2(b);
    fun2(d);
    return 0;
}

```

7、 What is the output of the following program? (6 points)

```

#include <iostream>
using namespace std;
int main()
{
    int a[3][4]={1,2,3,4,5,6,7,8,9,10,11,12};
    cout<<a[0][1]<<endl;
    cout<<*(a[1]+1)<<endl;
    cout<<*(*(a+2)+1)<<endl;
    return 0;
}

```

8、 What is the output of the following program? (6 points)

```
#include <iostream>
using namespace std;
int main()
{   int a[100],b[100],j,m;
    int sumA=0,sumB=0,sumDiff=0;
    cin>>m;
    for(j=0;j<m;j++)
    {   cin>>a[j]>>b[j];
        sumA=sumA+a[j];
        sumB=sumB+b[j];
        sumDiff=sumDiff+(a[j]-b[j]);}
    for(j=m-1;j>=0;j--)
    {cout<<a[j]<<' '<<b[j]<<' '<<a[j] - b[j] <<endl;}
    cout<<endl;
    cout<<sumA<<' '<<sumB<<' '<<sumDiff<<endl;
    return 0;
}
```

Input data:

```
3
11 15
19 14
4 2
```

III.Fill the vacant position in the program? (10 points)

The following program is a recursion, which is called the Towers of Hanoi. Please fill the vacant of the program.

```
#include <iostream>
#include <__1__>

using namespace __2__ ;

void __3__(int ,int ,int ,int);
```

```
int main()
{
    int circleCount;

    cout<<"Input number of circles:";
    cin>>circleCount;

    DoTowers(circleCount,1,2,3);

    return 0;
}

void DoTowers(int circleCount,int beginPeg,int auxPeg,int endPeg)

//This recursive function moves circleCount circles from beginPeg to endPge.
//all but one of the circles are moved from beginPeg to auxPeg,
//then the last circle is moved from beginPeg to endPeg,
//and then the circles are moved from auxPeg to endPeg.

{
    if( __4__ )
    {
        DoTowers( __5__ ,  beginPeg, __6__ , __7__ );
        cout<<setw(10)<<"Move circle"<<circleCount<<" from "<<beginPeg
            <<" to "<<endPeg<<endl;
        DoTowers( __8__ , auxPeg, __9__ , __10__ );
    }
}
```

IV. Programming Problems (26 points)

1. Write a function comb(int n, int k), implement the formula $C(n, k) = \frac{n!}{k!(n-k)!}$. (8 points)

2. Find (8 points)

Everyone who has used OFFICE software would be familiar with the "find" function. To simplify the problem, your task is to calculate the occurring time of a key word in a sentence.

Input:

There are several test cases in the input data. The first line contains the number of test cases. For each test case, there are two lines. The first line contains a sentence without any spaces in it. The second line contains the key word. All the input sentences or words consist of lowercase letters.

Output:

You should output the occurring time of the key word in the sentence.

Sample Input:

```
3
aaaa
aa
xxyyzz
xyz
goodluck
o
```

Sample Output:

```
3
0
2
```

3. Binary (10 points)

Binary system plays a crucial role in computer science. Given an integer n, we can easily write it in the binary form which is a 0-1 sequence. Your task is to calculate the number which is represented by that sequence in reverse order. For instance, the binary form of integer 6 is $(110)_2$, so the answer is 3 with the binary form $(011)_2$.

Input:

There are several test cases in the input data. The first line contains the number of test cases. For each case, there is an integer n in a separate line. ($1 \leq n \leq 10^5$)

Output:

Output the number which is represented by that sequence in reverse order.

Sample Input:

```
5
10
100
1000
10000
100000
```

Sample Output:

```
5
19
95
569
2755
```

I .Answer:

1	2	3	4	5	6	7	8
B	D	D	C	A	C	A	B

II .Answer:

1、 x=2.333 y=2.000 z=1.000	2、 The number is: one two three four
3、 x=10 y=7 x=7 y=10	4、 1 1 1 2 1 3
5、 2 2 4 4 5	6、 BDBB
7、 2 6 10	8、 4 2 2 19 14 15 11 15 -4 34 31 3

III. Answer:

- (1) iomani
- (2) std
- (3) DoTower
- (4) circleCount>0
- (5) circleCount-1
- (6) endPag
- (7) auxPag
- (8) circleCount-1
- (9) beginPag
- (10) endPag

1. Example:

```
#include <iostream>

using namespace std;

long fact(int n)
{
    if (n < 2) return 1;

    long f = 1;

    for (int i=2; i<=n; i++)

        f=f*i;

    return f;
}

long comb(int n, int k)
{
    if (n<0 || k<0 || k>n) return 0;

    return fact(n)/(fact(k)*fact(n-k));
}

int main()
{
    int n,k;

    cin >>n >>k;

    cout << comb(n,k)<<endl;

    return 0;
}
```

2. Example:

```
#include <iostream>
#include <string>
using namespace std;
int main()
{
    string str, sstr;
    int cases,i;
    cin >> cases;
    for (i=0; i<cases; i++)
    {
        cin >> str >> sstr;
        int count = 0;
        int loc = 0;
        while (loc < str.length())
        {
            loc = str.find(sstr, loc);
            if ( loc == string::npos)
                break;
            else
            {
                count++;
                loc++;
            }
        }
        cout<<count<<endl;
    }
    return 0;
}
```

3. Example:

```
#include <iostream>
using namespace std;
int main()
{
    int cases,i,j,k,l,a,b;
    int arr[20];
    cin >> cases;
    for (i=0; i<cases; i++)
    {
        cin >> a;
        j = 0;
        while (a > 1)
        {
            if (a%2 == 0)
                arr[j] = 0;
            else
                arr[j] = 1;
            a=a/2;
            j++;
        }
        if (a==1)
            arr[j] = 1;
        else
            j--;
        k = 0;
        b = 0;
        for(l=j; l>=0; l--)
        {
            b = b + (arr[l] << k);
            k++;
        }
        cout << b << endl;
    }
    //system("pause");
    return 0;
}
```